

Supplementary Material for “CTF-GS: Compact Foundation-Feature Fields for Open-Vocabulary 3D Gaussian Scene Understanding”

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I. OVERVIEW

This supplementary material records additional calibration, protocol controls, and qualitative examples for the main paper. The main paper keeps the narrative focused on the deployed compact field: rendered-view LERF-OVS grounding, OpenGaussian-style direct 3D object selection, VALA/OpenGaFF-8 ScanNet direct point querying, and frame-wise foundation-feature comparisons. The tables below provide the supporting controls behind those claims.

Unless otherwise stated, all LERF-OVS numbers are evaluated on the same four annotated scenes used in the main paper. Direct 3D object-selection controls render selected Gaussian primitives only for mask evaluation; the text query is applied before rendering.

II. RENDERED-VIEW BOUNDARY CALIBRATION

Table I expands the rendered-view mask calibration summarized in the main paper. The fixed threshold and peak-connected-component readout are label-free and use the heatmap argmax only to remove disconnected regions far from the queried peak.

III. DIRECT 3D PROTOCOL CONTROLS

Tables II–V document the registration and confidence controls used to understand direct 3D primitive querying. Multiview Primitive Registration is treated as a registered supervision and analysis bridge, while the main compact direct readout in the paper uses Gaussian-center primitive scores with fixed Support-Calibrated Primitive Readout and does not read a multiview-registration feature cache at inference.

TABLE I

LABEL-FREE RENDERED-MASK BOUNDARY CALIBRATION ON THE FIXED LERF-OVS CHECKPOINTS. THRESHOLD VARIANTS CHANGE ONLY THE PEAK-RELATIVE BINARY MASK THRESHOLD. THE FINAL VARIANT KEEPS THRESHOLD 0.60 AND ADDS PEAK-CONNECTED-COMPONENT FILTERING, WHICH REMOVES DISCONNECTED MASK REGIONS AWAY FROM THE QUERY HEATMAP MAXIMUM. LOCAcc IS MEASURED FROM THE HEATMAP ARGMAX BEFORE MASK EXTRACTION.

Readout	Fig.	Ramen	Tea.	Waldo	Macro
Threshold 0.50	0.431	0.586	0.549	0.411	0.494
Threshold 0.55	0.430	0.611	0.568	0.451	0.515
Threshold 0.60	0.424	0.620	0.576	0.477	0.524
Threshold 0.65	0.415	0.619	0.578	0.490	0.525
Threshold 0.70	0.395	0.596	0.556	0.477	0.506
Threshold 0.60 + peak comp.	0.513	0.625	0.618	0.527	0.571

TABLE II

MULTIVIEW PRIMITIVE REGISTRATION CONTROLS FOR LERF-OVS DIRECT 3D OBJECT SELECTION. EXCEPT FOR THE EXPLICITLY MARKED EARLIER RAW-CENTER VARIANT, RESULTS USE THE SAME FIXED TOP-2% PRIMITIVE SELECTOR. "REG." IS THE MEAN FRACTION OF GAUSSIANS RECEIVING REGISTERED RENDERED-VIEW FEATURES; UNREGISTERED PRIMITIVES USE THE DIRECT GAUSSIAN-CENTER FALLBACK UNLESS DISABLED BY THE ABLATION.

Variant	Selector / views	Reg.	mIoU	Acc@0.25
Raw Gaussian-center	top10%, earlier	–	0.080	0.093
Raw Gaussian-center	top2%, MPR protocol	–	0.012	0.009
MPR, cosine	top2%, all poses 24	0.385	0.312	0.547
MPR, softmax	top2%, all poses 24	0.385	0.342	0.555
MPR + voxel	top2%, train views 24	0.369	0.349	0.595
MPR + voxel	top2%, all poses 8	0.240	0.277	0.487
MPR + voxel	top2%, all poses 48	0.448	0.380	0.653
MPR + voxel	top2%, all poses 96	0.485	0.385	0.643
MPR + voxel, lower blend	top2%, all poses 96	0.485	0.385	0.631
MPR + voxel, w/o calibration	top10%, all poses 24	0.385	0.070	0.099
MPR + voxel, w/o depth check	top2%, all poses 24	0.972	0.347	0.615

TABLE III

MULTIVIEW-REGISTRATION-TO-FIELD CONSISTENCY FOR LERF-OVS DIRECT 3D OBJECT SELECTION. ALL VARIANTS USE THE SAME SOFTMAX-THRESHOLD SELECTOR FAMILY, 0.5% FLOOR, 1.8% CAP, AND LABEL-FREE COLOR-EDGE BOUNDARY CALIBRATION. RAW AND REGISTRATION-TO-FIELD VARIANTS REPORT PER-SCENE SWEEP-BEST THRESHOLDS; REGISTERED MPR REPORTS THE FIXED `THR0P25` PAPER RESULT. REGISTRATION-TO-FIELD DISTILLS CACHED REGISTERED SUMMARY FEATURES INTO THE COMPACT DIRECT FIELD AND EVALUATES THE DIRECT GAUSSIAN-CENTER READOUT WITH A 0.5 RESIDUAL BLEND TO THE ORIGINAL FIELD. THE CONFIDENCE-WEIGHTED VARIANT USES LABEL-FREE LOG VIEW-COUNT WEIGHTS FROM MULTIVIEW REGISTRATION.

Readout	Metric	Fig.	Ramen	Tea.	Waldo	Macro
Raw Gaussian-center	mIoU	0.001	0.006	0.158	0.019	0.046
MPR-to-field consistency	mIoU	0.488	0.438	0.510	0.212	0.412
MPR-to-field + conf. weights	mIoU	0.404	0.598	0.520	0.225	0.436
Registered MPR readout	mIoU	0.531	0.580	0.566	0.243	0.480
Raw Gaussian-center	Acc@0.25	0.000	0.000	0.237	0.045	0.071
MPR-to-field consistency	Acc@0.25	0.714	0.578	0.695	0.364	0.588
MPR-to-field + conf. weights	Acc@0.25	0.643	0.746	0.678	0.409	0.619
Registered MPR readout	Acc@0.25	0.786	0.746	0.763	0.409	0.676

TABLE IV

DIRECT3D MECHANISM AUDIT USING LABEL-FREE MULTIVIEW-REGISTRATION COVERAGE, PRIMITIVE TEACHER-SCORE CONFIDENCE, AND TEXT-MARGIN AMBIGUITY. SCENE RESULTS REPORT MEAN VALID REGISTERED VIEWS; TEACHER-BUCKET RESULTS REPORT MEAN TOP-1% PRIMITIVE TEXT SCORE; TEXT-MARGIN RESULTS REPORT SCORE MARGIN AGAINST THE STRONGEST COMPETING CATEGORY.

Group	Item	Support	Coverage/score	mIoU	Acc@0.25
Scene	figurines	56	9.5911	0.5309	0.7857
Scene	ramen	71	20.8716	0.5805	0.7465
Scene	teatime	59	13.0006	0.5662	0.7627
Scene	waldo_kitchen	22	5.7542	0.2429	0.4091
Teacher bucket	low	70	0.1800	0.4345	0.6143
Teacher bucket	mid	69	0.4257	0.5132	0.7101
Teacher bucket	high	69	0.6939	0.6358	0.8551
Text margin	ambiguous	70	-0.0882	0.4329	0.6286
Text margin	mixed	69	0.2451	0.5303	0.7246
Text margin	distinct	69	0.6074	0.6202	0.8261

TABLE V

QUERY-LEVEL AUDIT FOR LERF DIRECT 3D OBJECT SELECTION. CONFIDENCE INTERVALS ARE BOOTSTRAP INTERVALS OVER QUERY MASKS FOR THE FIXED `THR0P25` SELECTOR.

Scene	Queries	mIoU	Acc@0.25	Zero pred.
Figurines	56	0.531	0.786	0.071
Ramen	71	0.580	0.746	0.028
Teatime	59	0.566	0.763	0.017
Waldo Kitchen	22	0.243	0.409	0.182

TABLE VI

NEAREST-VIEW CACHED-TEACHER BASELINE ON LERF-OVS. EACH ANNOTATED TARGET FRAME USES THE CLOSEST CACHED RADIO FEATURE MAP BY CAMERA-CENTER DISTANCE, EXCLUDING THE TARGET FRAME ITSELF; THE FEATURE MAP IS NOT WARPED INTO THE TARGET CAMERA.

Scene	LocAcc	mIoU	N	Dist.
Figurines	0.1786	0.0755	56	0.2831
Ramen	0.2817	0.1740	71	0.5443
Teatime	0.3559	0.2114	59	0.4266
Waldo Kitchen	0.2727	0.1570	22	0.5787
Macro	0.2722	0.1545	–	0.4582

TABLE VII

PER-GAUSSIAN 1280-D EXPLICIT RADIO-MEMORY BASELINE ON LERF-OVS. CACHED TEACHER FEATURES ARE REGISTERED TO GAUSSIAN CENTERS, STORED AS FP16 VECTORS, RENDERED TO ANNOTATED VIEWS, AND EVALUATED WITH THE SAME FROZEN SIGLIP2 SCORER.

Scene	LocAcc	mIoU	Reg. frac.	Storage MiB
Figurines	0.6429	0.2729	0.1123	412.1
Ramen	0.6056	0.4333	0.2835	934.3
Teatime	0.5085	0.3541	0.2647	1123.4
Waldo Kitchen	0.5000	0.2125	0.1475	1688.8
Macro	0.5642	0.3182	0.2020	1039.7

IV. FEATURE-MEMORY AND COMPRESSION CONTROLS

Tables VI–IX compare the compact field against simple feature-memory alternatives and relate feature reconstruction quality to downstream relevance. These controls support the central design choice of reconstructing compact foundation-space features rather than storing a raw high-dimensional teacher vector per primitive.

V. BOUNDARY AND GEOMETRY CONTROLS

Tables X and XI record boundary-oriented metrics for the SAM3-box boundary readout. This readout is reported as a boundary-assisted control rather than the core compact direct readout.

VI. TRAINING AUDITABILITY

Table XII records checks for the training entry point. The paper artifact manifest and checksum file provide the corresponding source-level provenance for tables, figures, and result registries.

TABLE VIII

COMPRESSION RATIO VERSUS DOWNSTREAM mIoU. SAVING RATIO COMPARES DIRECT PER-GAUSSIAN 1280-D FP16 TEACHER-FEATURE STORAGE WITH THE COMPACT CHECKPOINT FOOTPRINT; DOWNSTREAM METRICS USE THE FROZEN RENDERED-GROUNDING AND DIRECT3D READOUTS.

Scene	Saving	Rendered mIoU	Direct3D mIoU	Direct3D Acc@0.25
Figurines	1.74×	0.4244	0.5309	0.7857
Ramen	3.00×	0.6201	0.5805	0.7465
Teatime	3.32×	0.5760	0.5662	0.7627
Waldo Kitchen	4.04×	0.4769	0.2429	0.4091

TABLE IX

FEATURE RECONSTRUCTION ERROR VERSUS TEXT RELEVANCE ERROR. FEATURE ERROR IS 1— BEST VALIDATION DECODED COSINE FROM THE SCENE TRAINING LOG; TEXT RELEVANCE ERRORS ARE DERIVED FROM THE FROZEN RENDERED-GROUNDING READOUT.

Scene	Feature err.	mIoU err.	LocAcc err.	Best cosine
Figurines	0.2382	0.5756	0.1786	0.7618
Ramen	0.1624	0.3799	0.0986	0.8376
Teatime	0.2005	0.4240	0.1017	0.7995
Waldo Kitchen	0.2209	0.5231	0.1364	0.7791

TABLE X

MEASURED BOUNDARY-ERROR READOUT FOR THE STRICT SAM3-BOX DIRECT3D ROW. BOUNDARY ERROR IS 1—BOUNDARY-F AND TRIMAP ERROR IS 1—TRIMAP IoU.

Scene	mIoU	Boundary-F	Boundary err.	Trimap IoU	Trimap err.
Figurines	0.6136	0.7116	0.2884	0.3986	0.6014
Ramen	0.6409	0.7713	0.2287	0.4400	0.5600
Teatime	0.6130	0.7255	0.2745	0.4626	0.5374
Waldo Kitchen	0.4142	0.4638	0.5362	0.2820	0.7180

TABLE XI

ALPHA/DEPTH DISCONTINUITY ALIGNMENT COVERAGE FOR THE STRICT SAM3-BOX DIRECT3D ROW. GEOMETRY RECORDS COUNT PER-QUERY ALPHA/DEPTH EDGE METRICS AND OVERLAYS.

Scene	Queries	Geometry records	mIoU	Boundary-F
Figurines	56	56	0.6136	0.7116
Ramen	71	71	0.6409	0.7713
Teatime	59	59	0.6130	0.7255
Waldo Kitchen	22	22	0.4142	0.4638

TABLE XII

AUDITABILITY CHECKS FOR THE TRAINING ENTRY POINT. STATUS ‘RISK’ MEANS THE GUARD EXISTS BUT NEEDS RELEASE HARDENING.

Check	Status	Severity	Evidence
script_size	pass	medium	3735 lines; threshold 4000
run_manifest	pass	high	present
split_resolution	pass	high	train/val split tokens present
trusted_checkpoint_io	pass	high	load_trusted_checkpoint referenced
training_lock	pass	medium	lock acquisition/release present
raw_tensor_load_sites	pass	medium	1 raw torch.load site behind or replaced by load_training_tensor_cache

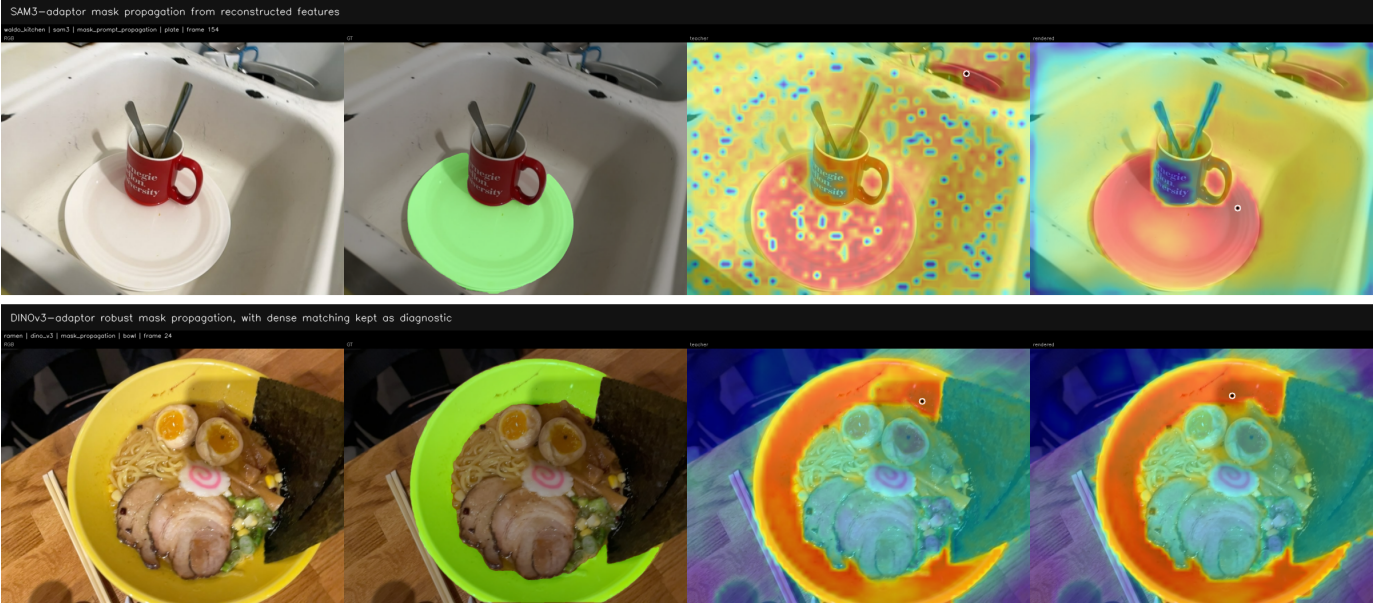


Fig. 1. Additional frozen-head qualitative examples. The SAM3-adaptor and DINOv3-adaptor panels are driven by reconstructed CTF-GS features rather than an official RGB SAM decoder.

VII. ADDITIONAL QUALITATIVE RESULTS

Figure 1 shows frozen-head adaptor examples. These examples are useful for interpreting the frame-wise foundation-feature comparison, but the main paper keeps the LERF qualitative figure focused on the cleaner 2D/3D open-vocabulary query taxonomy.

Figure 2 shows the registered multiview direct-3D control. It is included as an analysis readout because Multiview Primitive Registration transfers registered multiview support to primitives, whereas the main compact direct readout uses the deployed compact Gaussian-center field.

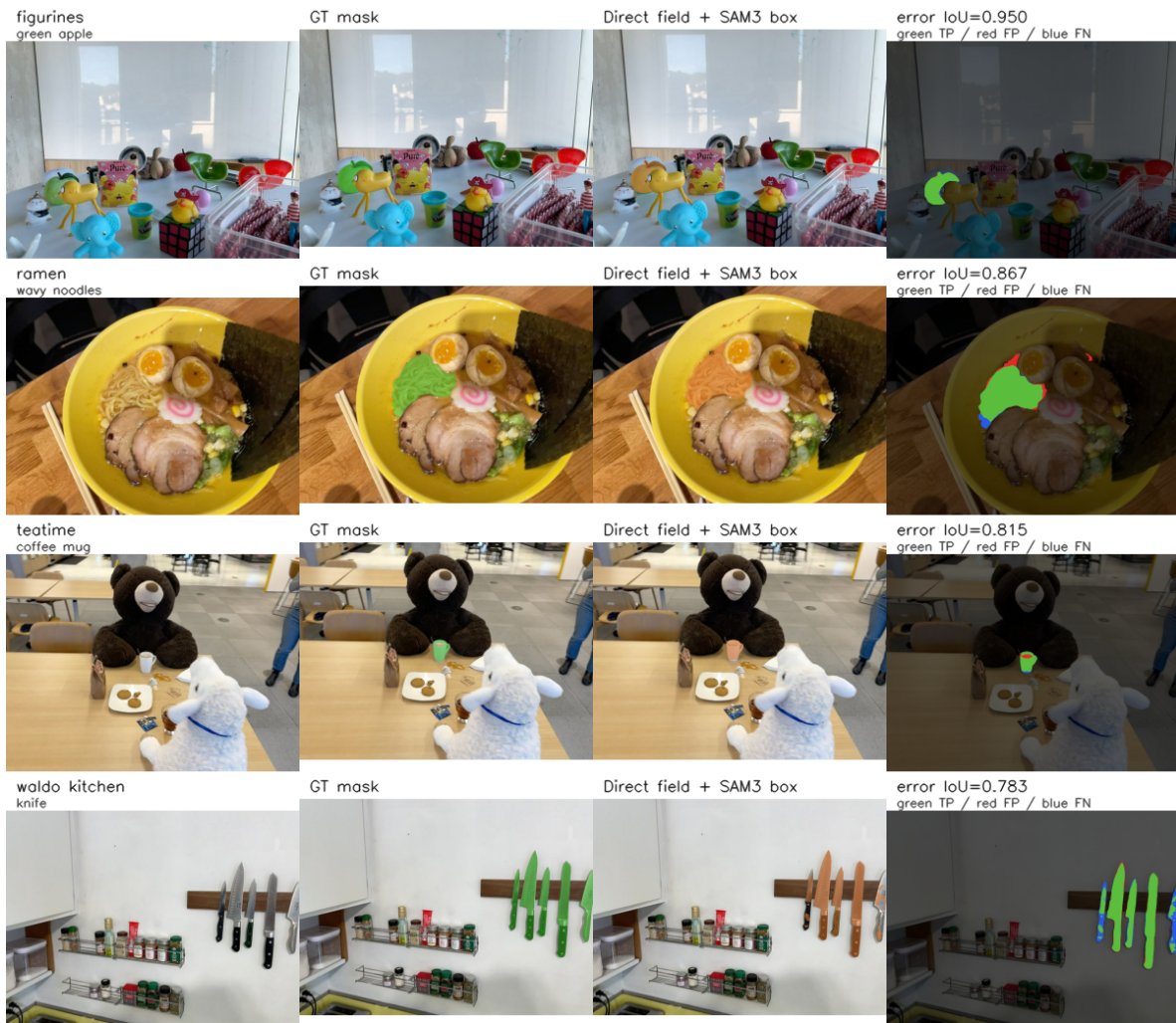


Fig. 2. Registered multiview direct-3D control examples. Multiview Primitive Registration is used as an analysis and training bridge, not as the main compact-field inference cache.

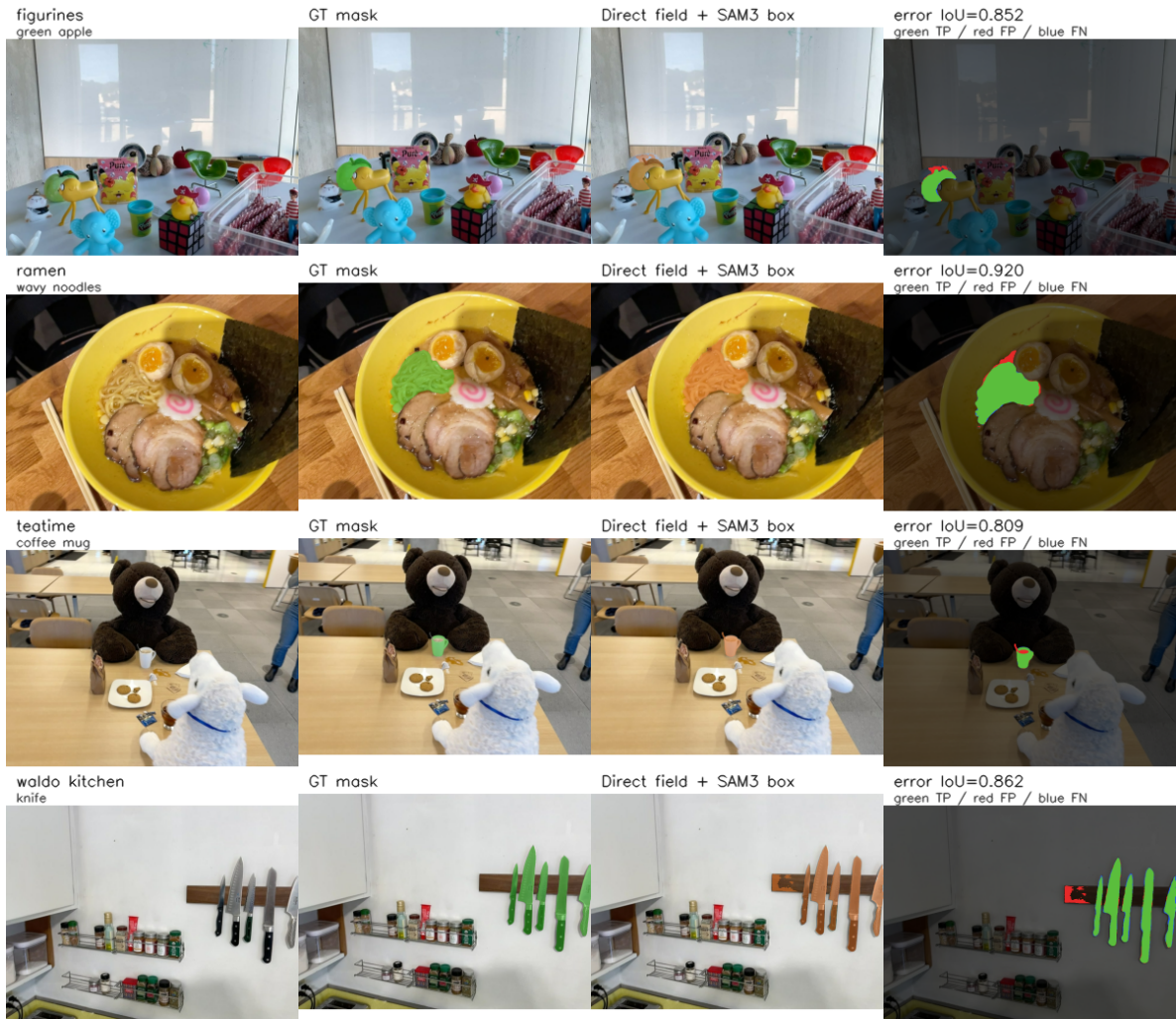
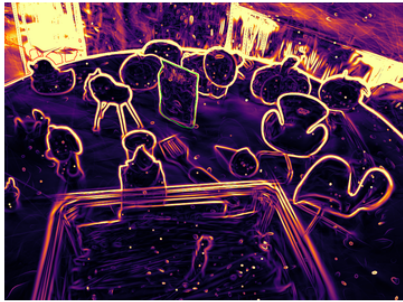


Fig. 3. Boundary-assisted direct-3D examples using the SAM3-box control. This variant illustrates a boundary readout after compact primitive selection and is separate from the strict compact one-map ablation.

1. Figurines / pikachu | IoU 0.00, BE 1.00



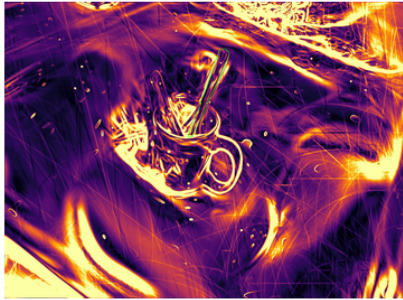
2. Ramen / napkin | IoU 0.00, BE 1.00



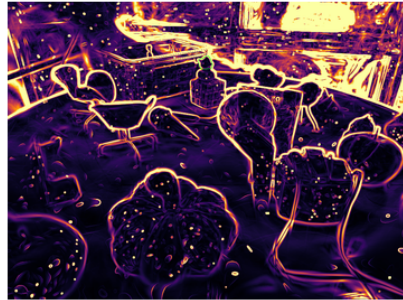
3. Teatime / yellow pouf | IoU 0.00, BE 1.00



4. Waldo Kitchen / knife | IoU 0.00, BE 1.00



5. Figurines / pirate hat | IoU 0.00, BE 1.00



6. Figurines / pirate hat | IoU 0.00, BE 1.00

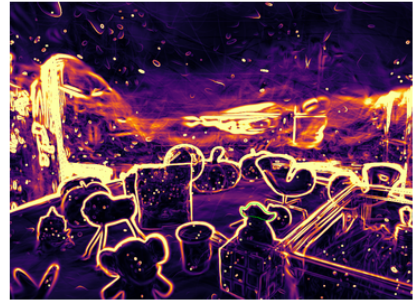


Fig. 4. Alpha/depth boundary cases used for geometric boundary analysis. These cases expose failure modes where low object support or depth discontinuities limit mask overlap.